

KARTHIK AKULA VINOD

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Professional Summary

Experienced Data Engineer with over 3 years of experience in building and optimizing data pipelines and ETL processes. Skilled in working with large datasets, designing data workflows, and improving data system performance. Proficient in using SQL, Python, and cloud platforms like AWS and Google Cloud to deliver efficient data solutions that support business decisions. Strong background in data management and collaborating with cross-functional teams to automate processes and enhance data-driven insights.

Skills

- **Programming Languages:** Python, SQL, Java
- **Data Technologies:** Apache Spark, Hadoop, Kafka
- **Cloud Platforms:** AWS (S3, Redshift, Lambda), Google Cloud (BigQuery, Dataflow)
- **Databases:** MySQL, MongoDB
- **Data Tools:** Apache Airflow, Talend, SSIS
- **Data Visualization:** Tableau, Power BI
- **Methodologies:** Agile, Scrum

Professional Experience

Data Engineer

NYU Langone, New York, NY | 02/2024 – Present

- Developed and optimized ETL pipelines using Python and SQL to integrate data from multiple sources into Amazon Redshift.
- Managed data lakes on AWS S3 to handle large datasets for analysis and machine learning.
- Created real-time data streaming solutions with Apache Kafka, reducing data processing latency by 30%.
- Worked with cross-functional teams to ensure data quality and provide clean data for business reporting and analysis.
- Automated the deployment process using Jenkins and Docker, reducing deployment time by 25%.
- Developed a real-time fraud detection pipeline, decreasing potential losses by 20%.

Data Engineer

Virtusa Consulting Services, India | 05/2021 - 07/2022

- Designed and implemented complex data pipelines with Apache Spark and Hadoop to process large-scale data in batch and real-time modes.
- Automated data processing workflows using Apache Airflow, reducing manual intervention by 50%.
- Integrated data from on-premise and cloud sources into Snowflake and Google BigQuery, improving query performance by 30%.
- Developed data models to support analytics and reporting needs for finance and marketing departments.
- Led a team to implement a scalable data lake on AWS S3, enabling real-time analytics.
- Enhanced data integration and reporting, improving decision-making efficiency by 35%.

Data Analyst Intern

Dentsu, India | 08/2020 - 02/2021

- Assisted in developing data pipelines using SQL and Python to support marketing analytics initiatives.
- Collaborated with cross-functional teams to define data requirements and improve data quality.
- Created interactive dashboards using Tableau to visualize key marketing KPIs and campaign performance.

Projects

- **Retail Customer Behavior Analysis**
Objective: Analyzed transaction data to identify buying patterns and trends.
Outcome: Increased sales by 20% and improved customer retention by 15%.
- **Smart City Traffic Flow Analysis**
Objective: Optimized traffic flow and reduced congestion using BigQuery and Tableau.
Outcome: Reduced congestion by 25% and improved travel times by 15%.

Certifications

- Oracle Certified Associate, Java SE 8 Programmer
- AWS Partner: Accreditation (Business) (Digital)
- AWS Partner: Cloud Economics Accreditation

Education

- **Master of Science (M.S.) in Management Information Systems**
Northern Illinois University | May 2024
- **Bachelor of Technology (B.Tech.) in Computer Science Engineering**
Geethanjali College of Engineering and Technology | October 2020